Session Objectives

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At the conclusion of the session, the student should be able:

1. Describe the major functions of the kidneys.

2. Define & describe the gross structures and their interrelationships: renal pelvis, renal pyramids, renal medulla (inner and outer zones), renal cortex, and papilla.

3. Define & describe the components of the glomerular filtration barrier

4. Define & describe the components of the nephron-collecting duct system and their interrelationships.

5. Define & describe the juxtaglomerular apparatus and its role in homeostasis.

Functional Anatomy of the Kidney

- The kidneys serve three essential functions:
  - Filtration: Removes metabolic products & toxins from the blood & excretes them through the urine.
  - Regulation of fluid status, electrolyte balance & acid-base balance.
  - Production or activation of hormones involved in erythropoiesis, Ca^{++} metabolism, & regulation of blood pressure & flow.
- Paired, bean-shaped structures that lie behind the peritoneum on each side of the vertebral column (L12 to L3).
- Comprise less than 0.5% of the TBW
  - Males: 125-170 g each
  - Females: 115-155 g each.
- Hilus: Site of entry for the renal artery & nerves & exit for the renal vein.
- Renal artery: Enters the hilus and divides into anterior and posterior branches, which give rise to the arcuate and then interlobar arteries.
- Arcuate arteries:
  - Skirt the corticomedullary junction
  - Branch into ascending interlobular arteries → afferent arterioles.
- Renal sinus:
  - Includes the renal pelvis & the major and minor calyces (urine-filled spaces)
  - Surrounded by renal parenchyma except where it connects with the upper end of the ureter.

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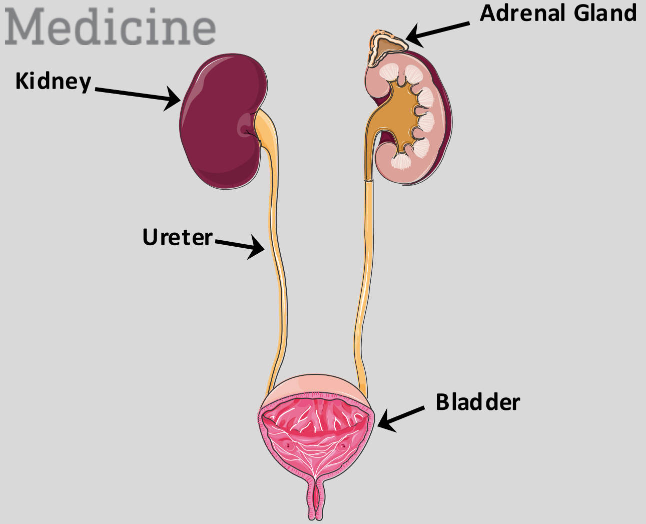

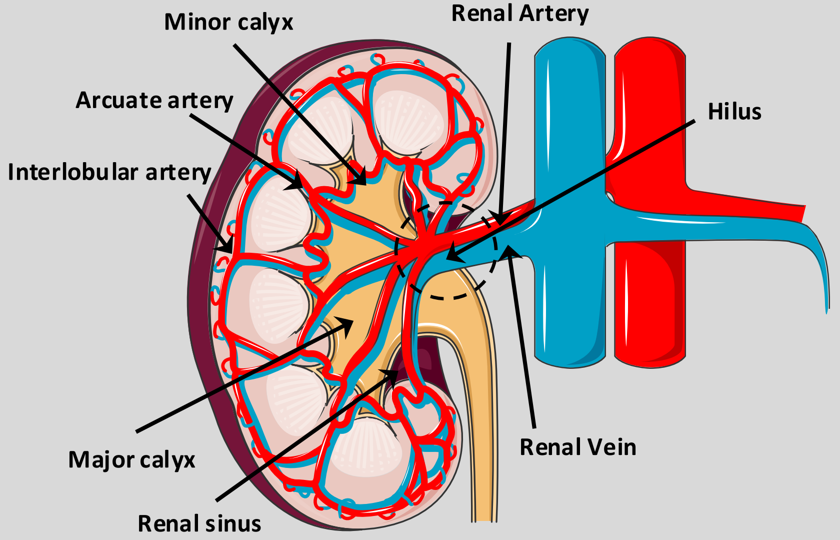

Functional Anatomy of the Kidney

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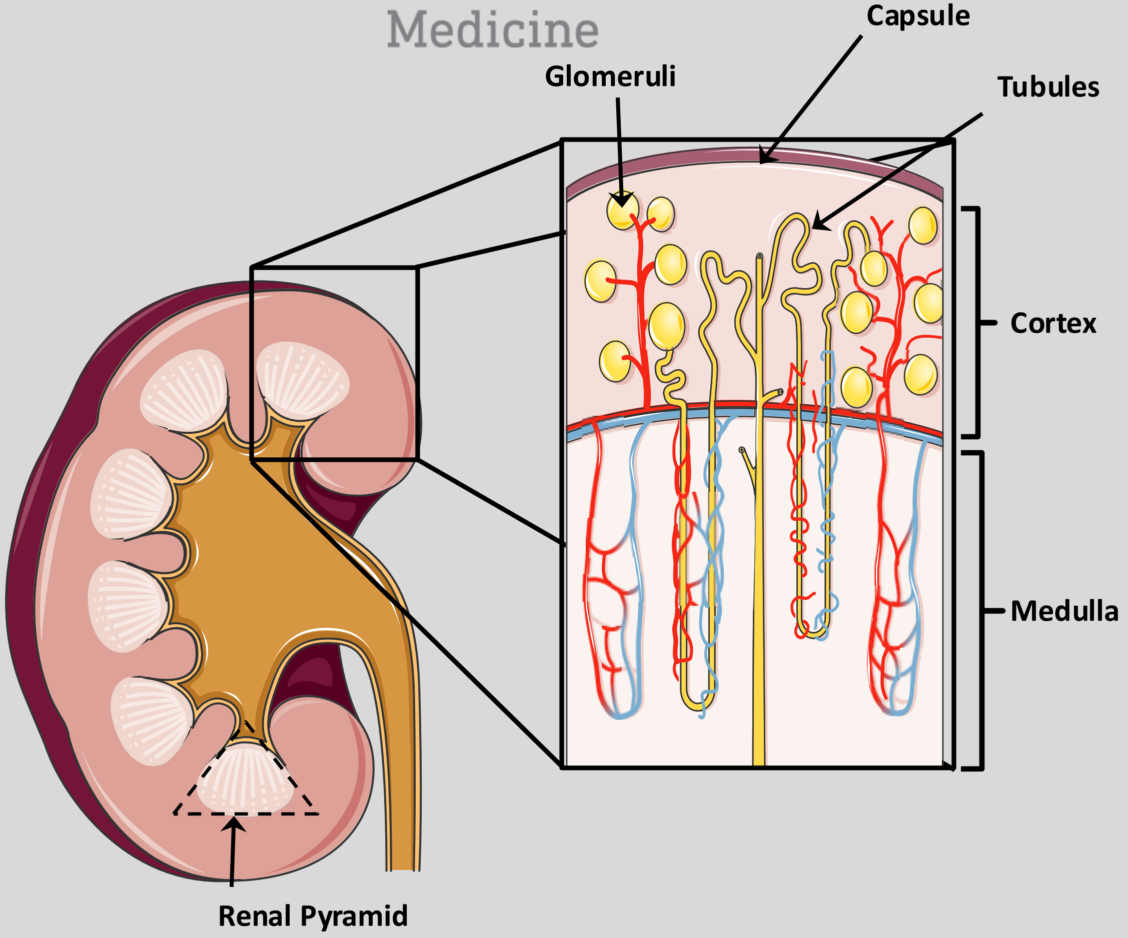
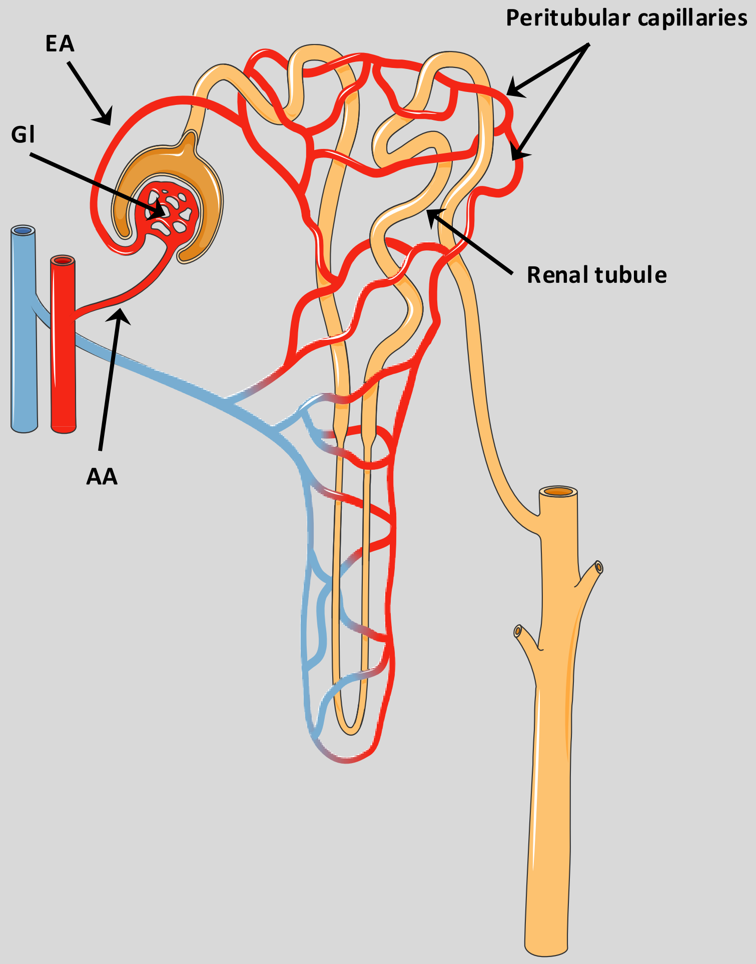
- **A fibrous capsule covers each kidney.**
- **The renal capsule reflects into the sinus at the hilus**
- **Its inner layers line the sinus**
- **Outer layers anchor to the blood vessels and renal pelvis.**
- **Kidney cross-section reveals two visible layers:**
  - **Cortex:**
    - **Granular outer region due to the presence of glomeruli & highly convoluted tubules**
  - **Medulla:**
    - **A darker inner region.**
    - **Lacks glomeruli**
    - **Parallel arrangement of tubules and small blood vessels.**
    - **Subdivided into 8-18 conical renal pyramids**
    - **Renal Pyramids :**
      - **Base faces the cortical-medullary border**
      - **Tip terminates in the renal pelvis.**



Vasculature of the Kidney

- The kidneys have a very high blood flow and complex & unique vasculature
- They receive ~ 20% of the cardiac output.
- High blood flow provides the blood plasma necessary for forming <u>ultrafiltrate</u>
- Renal circulation:
    - Afferent arteriole (AA): High-resistance arteriole bringing blood into the glomeruli.
    - Glomerulus (Gl): High-pressure capillary network for filtration
    - Efferent arteriole (EF): 2^{nd} high-resistance arteriole
    - Peritubular capillaries: Low-pressure capillary network surrounding the renal tubules
- Juxtamedullary glomeruli:
    - Located at or near the medullary & cortical junction.
    - Efferent arterioles of these nephrons descend into the renal papillae to form hairpin-shaped vessels called the <u>vasa recta</u>
    - Vasa Recta: The capillary network for tubules in the medulla.
- ~90% of the blood entering the kidney perfuses superficial GL and cortex
- ~10% perfuses juxtamedullary GL and medulla.
- Lymph vessels:
    - Drain cortical ISF & contain renal hormones (e.g., erythropoietin)
    - Leave the kidney by following the arteries toward the hilus.
    - Absent from the renal medulla



The Nephron

- The functional unit of the kidney is the nephron (800,000 to 1,200,000/kidney)
- Consists of a glomerulus and a tubule.
- Each nephron is an independent entity until the point at which its initial collecting tubule merges with another tubule.
- 2 populations of nephrons w/i the renal cortex:
  - Superficial nephrons: Have <u>short loops</u> extending to the boundary between outer and inner medulla.
  - Juxtamedullary nephrons: Play a special role in the production of concentrated urine with <u>long loops</u> that extend into medulla.
- Renal corpuscle:
  - The site of formation of the glomerular filtrate, comprises of the following:
    - Glomerulus: Cluster of capillaries from which the plasma filtrate originates.
    - Tubule: Epithelial structure consisting of subdivisions, designed to convert the filtrate into urine.
    - Bowman's capsule (glomerular capsule):
      - Epithelial structure surrounding the glomerulus
      - Contains Bowman's space
      - Contiguous with the lumen of the tubule.
      - Where the filtrate passes from the vascular into the tubule system.
- Bowman’s space:
  - The space between parietal cells & glomerular capillaries

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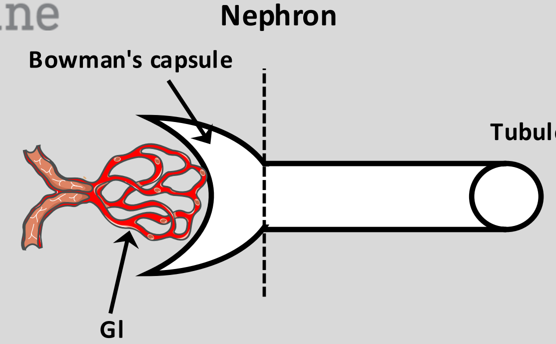
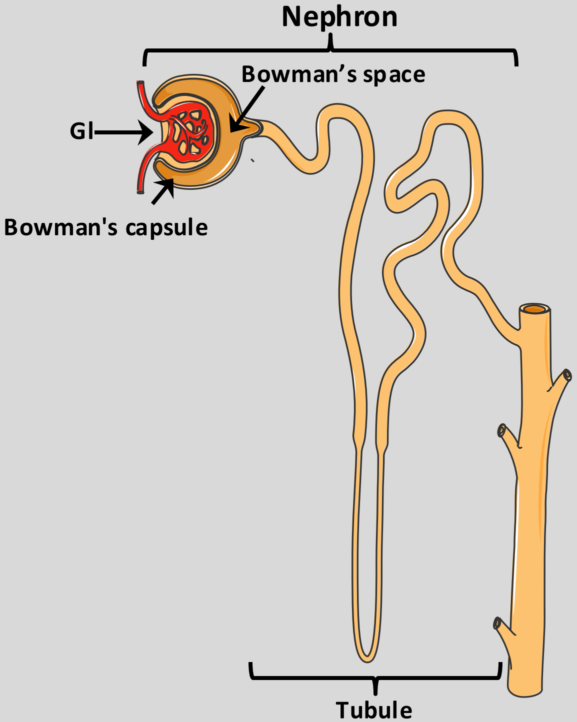
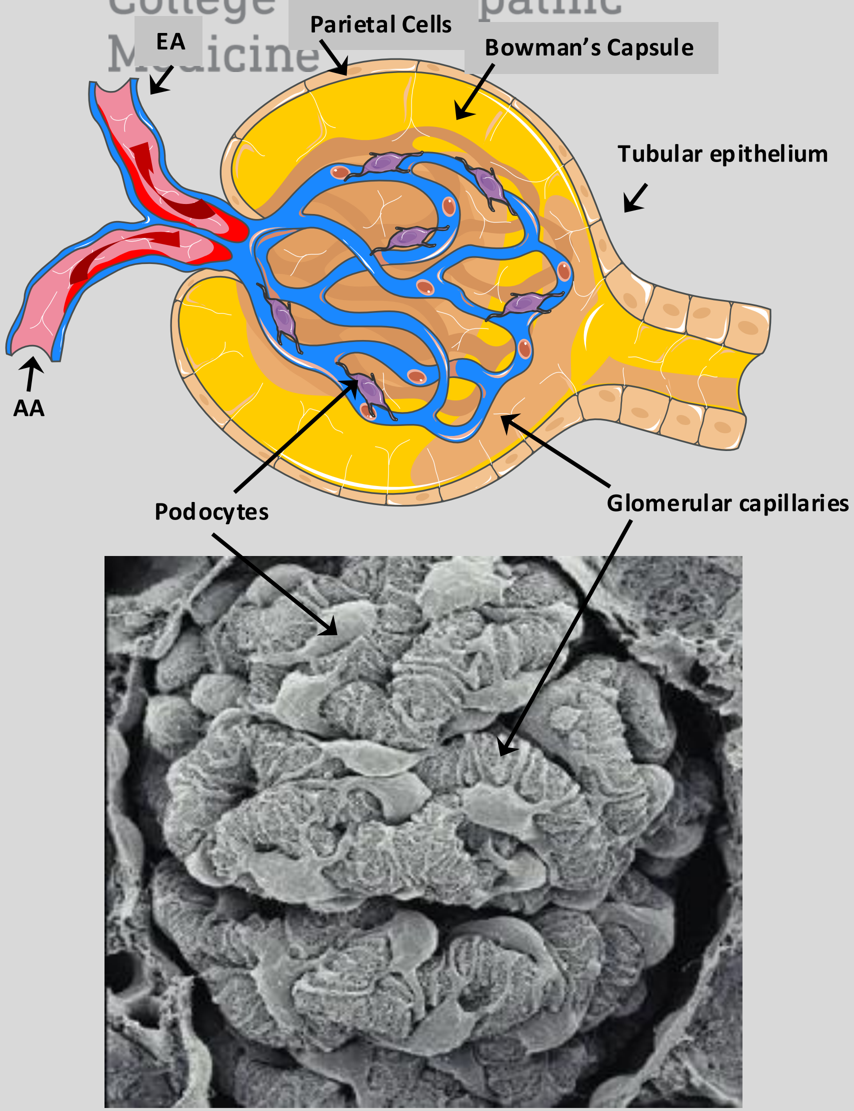




The Glomerulus & Bowman’s Capsule

Development:

- The ureteric bud gives rise to the urinary system from the collecting duct to the ureters
- The tubular epithelium between Bowman's capsule & the connecting ducts arise from condensed mesenchymal cells.
- The proximal end of tubular epithelium attaches to the arterial vascular bundle that develops into the glomerular capillary tuft.
- Thinning of the epithelium on one blind end of tubule leads to the emergence of the parietal layer of Bowman's capsule.
- In contrast, the opposite visceral layer thickens and attaches to the glomerular capillaries.
- These visceral epithelial cells later fold and develop into podocytes.
- Podocytes:
    - Modified epithelial cells
    - Represent the visceral layer of Bowman's capsule.
    - Podocytes are continuous with the parietal layer of Bowman's capsule.
- Ultrafiltrate drains into Bowman's space and flows into the proximal tubule at the urinary pole of the renal corpuscle.



The Glomerular Filtration Barrier

- In the mature kidney, foot processes of the podocytes cover the glomerular capillaries.
- Glomerular filtration barrier is between the glomerular capillary lumen and Bowman's space
  - Glycocalyx:
    - Negatively charged glycosaminoglycans
    - Prevents leakage of large negatively charged macromolecules.
  - Endothelial cells:
    - Surrounded by the glomerular basement membrane and a layer of podocyte foot processes.
    - Contain large fenestrations, 70-nm holes
    - No restriction to movement of H_2O, small solutes, small proteins out of the capillary
  - Basement membrane:
    - Located between endothelial cells (inner) and podocyte (outer) foot processes
    - Three layers
      - Lamina rara interna: Inner thin layer
      - lamina densa: Thick layer
      - Lamina rara externa: Outer thin layer
    - Contains heparan sulfate proteoglycans
      - Restricts large, negatively charged solutes

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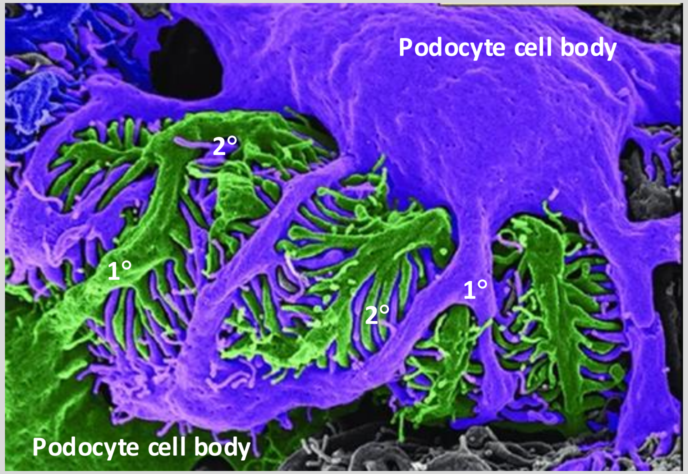
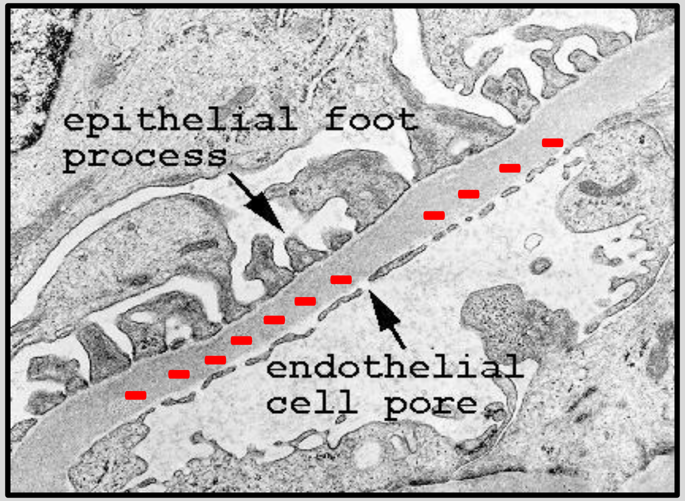
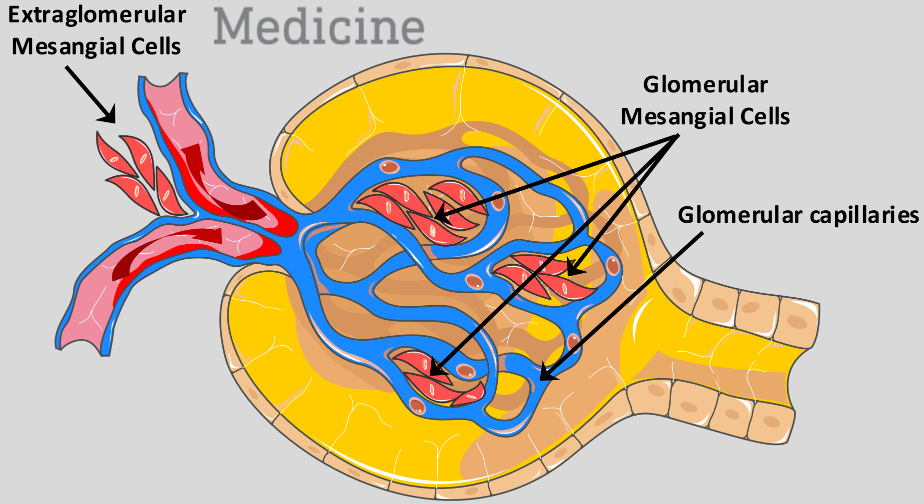


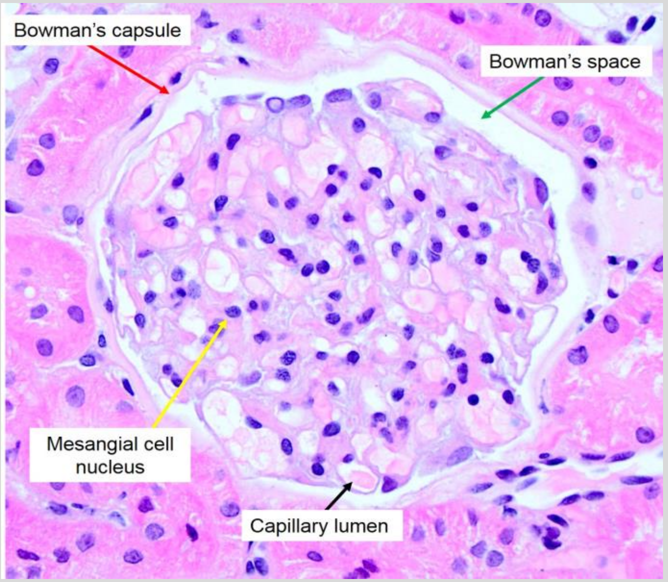


The Glomerular Filtration Barrier

Mesangial cells:
- Originate from embryonic mesenchymal cells
- Role is to maintain structural integrity of the glomerular microvasculature & ECM homeostasis.
- Possess contractile properties of smooth muscle cells.
- Influence glomerular filtration rate (GFR) by regulating blood flow through the glomerular capillaries or by altering the capillary surface area.
- Provide structural support for the glomerular capillaries
- Secrete extracellular matrix
- Exhibit phagocytic activity by removing macromolecules from the mesangium.
- Secrete prostaglandins & proinflammatory cytokines.

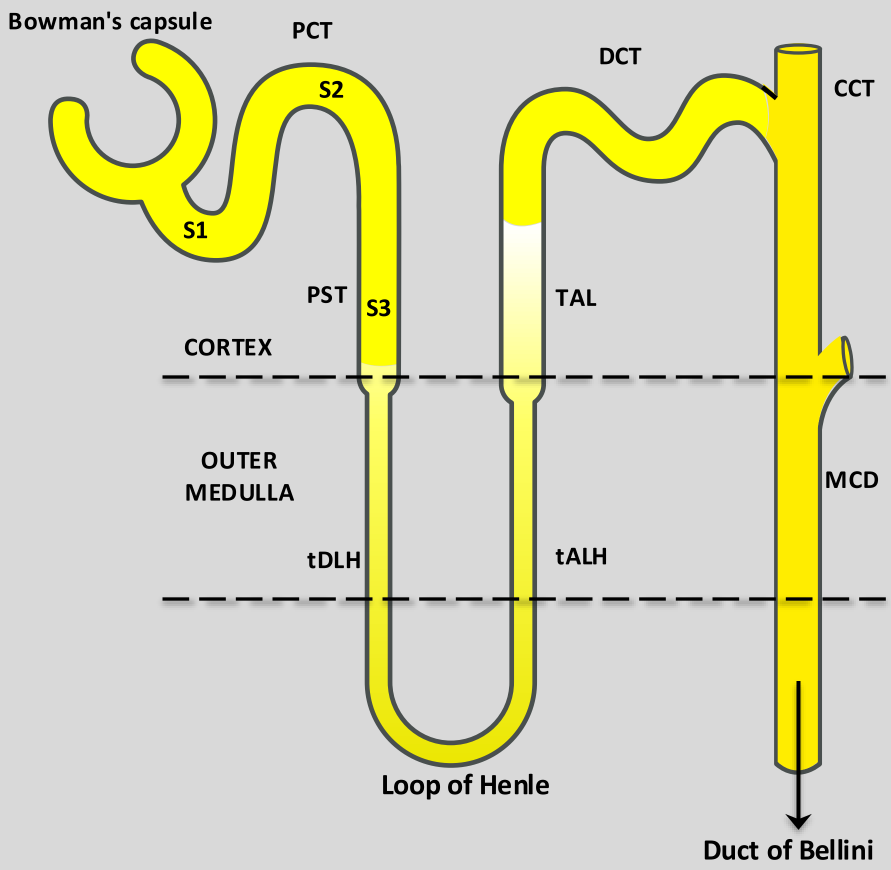
Extraglomerular Mesangial cells:
- Found outside the GL (between the afferent and efferent arterioles).
- Interact with *Macula Densa* (MD; specialized tubule epithelial cells) & granular cells of the afferent arterioles to form the Juxtaglomerular Apparatus (JGA).
- One component of the tubuloglomerular feedback (TGF) mechanism.





Tubular Components of the Nephron

- Embryologically derived from the ureteric bud & metanephric blastema.
- An epithelial structure designed to convert ultrafiltrate → urine.
- Bowman's capsule (glomerular capsule):
  - Surrounds the glomerulus & contains Bowman's space
  - Contiguous with the lumen of the tubule.
- Proximal tubules are subdivided into 3 segments based on their appearance:
  - Proximal Convoluted Tubule (PCT, S1 & S2):
    - Cells have apical microvilli and basolateral membrane (brush border)
    - Rich supply of mitochondria
  - Proximal straight tubule (PST, S2 & S3):
    - Cells have diminished ultrastructural complexity.
- Thin descending limb (tDLH) & thin ascending limb (tALH):
  - Less complex and flatter cells than PCT & PST
- Thick ascending limb of the loop of Henle (TAL):
  - Cells have tall interdigitations and numerous mitochondria within invaginated basolateral membranes
- Distal convoluted tubule (DCT):
  - Cells are similar to the thick ascending limb
- Cortical collecting tubule (CCT) & medullary collecting duct (MCD):
  - Composed of INTERCALATED and PRINCIPAL cells



Tubular Components of the Nephron

Proximal tubule:

- Reabsorbs 67% of the filtered H_2O & electrolytes (isosmotic)
- Reabsorbs all glucose, proteins, AA & most of HCO_3^- in the ultrafiltrate
- Highly permeable to H_2O (due to expression of AQP1) in the apical & basolateral membranes

Thin descending limb of the Loop of Henle:

- Reabsorbs 15% of the filtered H_2O (mediated by AQP1).

Thin ascending limb of the Loop of Henle:

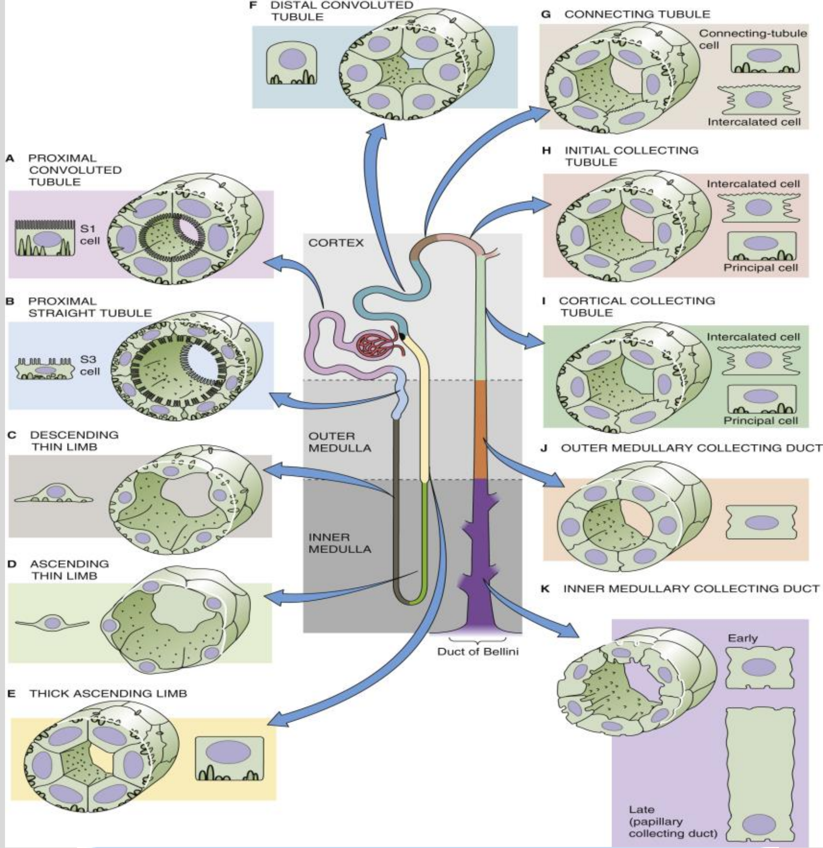
- Reabsorbs NaCl by a passive mechanism.
- H_2O reabsorption in the descending thin limb increases [NaCl] in the tubule fluid.
- NaCl-rich fluid moves toward the cortex
- NaCl diffuses out of the tubule lumen & into the medullary interstitial fluid.

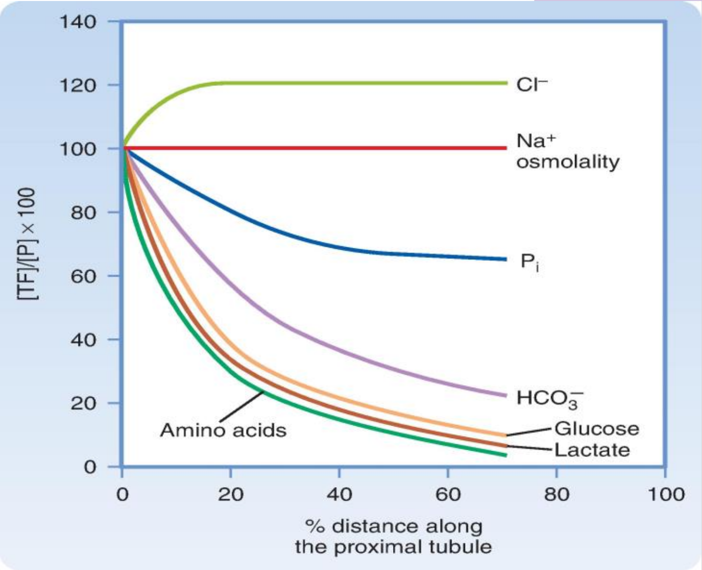
Thick ascending limb of the Loop of Henle:

- Impermeable to water.
- Reabsorbs 25% of filtered NaCl.
- K^+ is secreted & reabsorbed.
- Mg^{++} & Ca^{++} are reabsorbed paracellularly.
- Remainder of HCO_3^- is reabsorbed.

Distal convoluted tubule & Collecting ducts:

- Reabsorbs ~ 8% of the filtered NaCl
- Secrete variable amounts of K^+ & H^+
- Reabsorbs ≈ 8-17% of H_2O.





H_2O & Solute Permeability in the Nephron

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- Kidneys produce a range of urine osmolalities (concentrated (high mOsm)$\rightarrow$ dilute (low mOsm)).
- Nephron segments have differential permeabilities to solute & H_2O.

Proximal Tubule:
- H_2O reabsorption: Paracellular
- Solute reabsorption: Transcellular & Paracellular

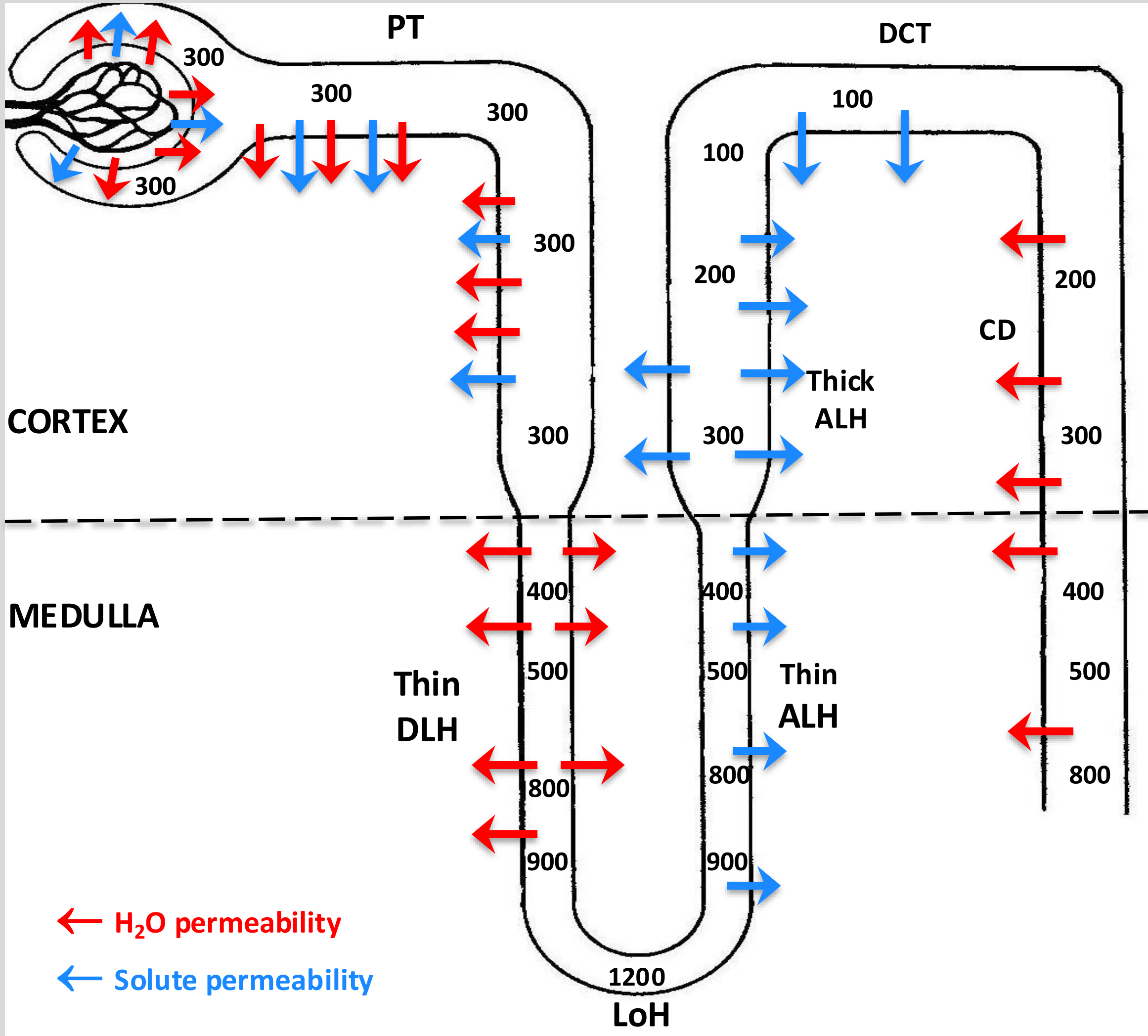
Thin Descending Limb of the Loop of Henle:
- H_2O reabsorption: Paracellular
- Solute reabsorption: None

Thin Ascending Limb of the Loop of Henle:
- H_2O reabsorption: None
- Solute reabsorption: Paracellular

Thick Ascending Limb of the Loop of Henle:
- H_2O reabsorption: None
- Solute reabsorption: Transcellular & Paracellular

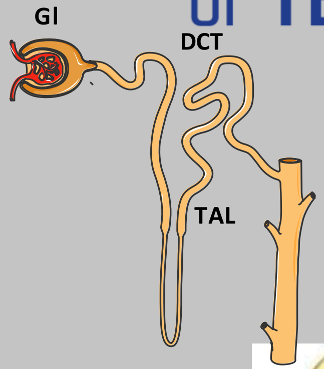
Distal Tubule:
- H_2O reabsorption: None
- Solute reabsorption: Paracellular

Collecting Duct:
- H_2O reabsorption: Transcellular
- Solute reabsorption: Transcellular & Paracellular

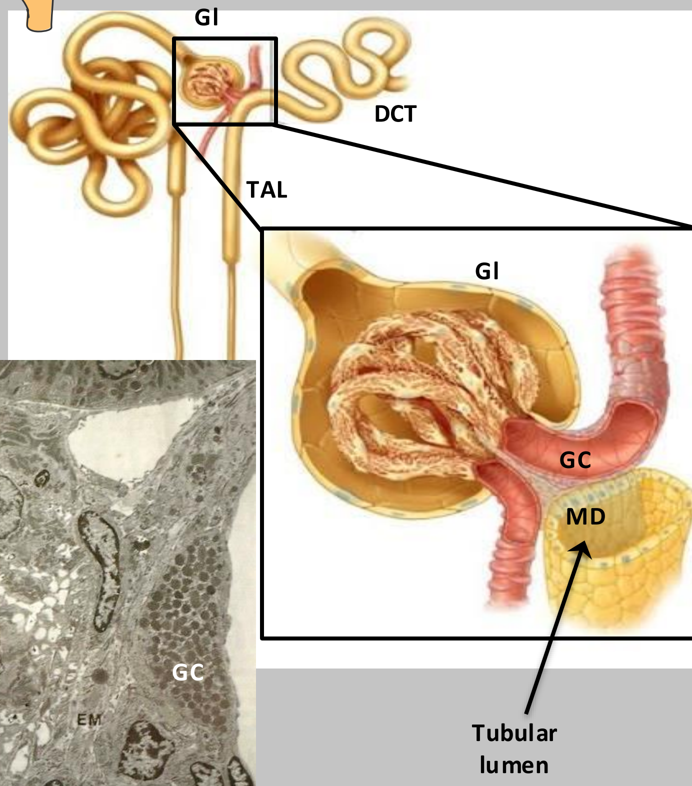


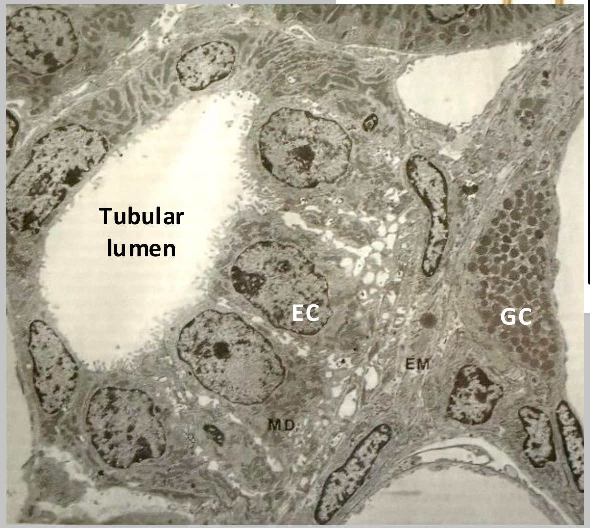
The Juxtaglomerular Apparatus

- Consists of extraglomerular mesangial cells, the *macula densa*, and granular cells (GC).
- *Macula densa*:
  - A patch of specialized tubule epithelial cells (EC) at the transition between the TAL and DCT
  - Contacts its “own” Gl.
  - Cells have large nuclei and are closely packed
- Granular cells (GC) of the afferent arterioles (juxtaglomerular or epithelioid cells) are specialized smooth-muscle cells that produce, store, and release the enzyme <u>renin.</u>
- JGA is part of a feedback mechanism that directly regulates RBF and filtration rate & indirectly modulates Na^+ balance & systemic BP.
- Two regulatory roles:
  - Tubuloglomerular feedback (TGF): Regulation of GFR depending on H_2O/NaCl reaching a *macula densa* increases.
  - Control of RBF: Detects changes in the pressure of the renal artery via baroreceptors in AA, directing granular cells to modulate renin (rē-nən; ree-nen) release into circulation and regulating the renin-angiotensin-aldosterone system (RAAS) .



Macula (spot) *densa* (dense)





Renal Innervation

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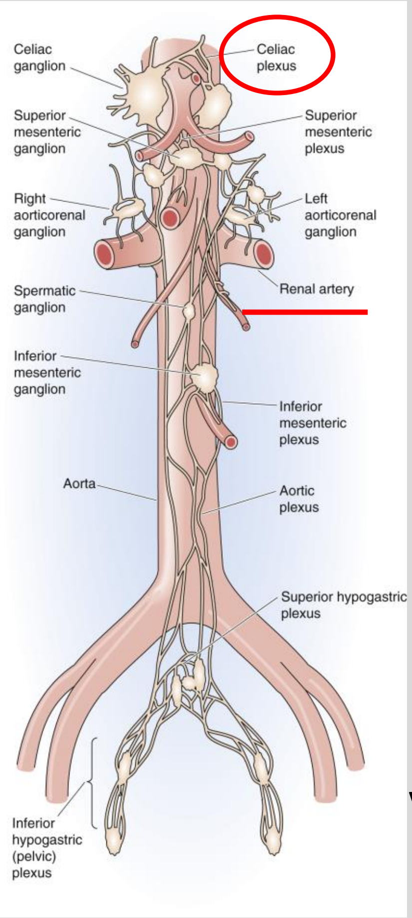
Efferent Innervation:

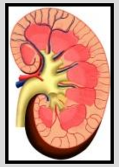
- Innervated by the sympathetic nervous system (SNS).
- One of the organs that has single innervation (Sweat glands, circulatory system, adrenal medulla, pancreas, liver, adipose tissues, kidney)
- Lack parasympathetic innervation.
- Renal SNS originates from the celiac plexus & follows the arterial vessels into the kidney
- Releases norepinephrine (NE) into the loose connective tissue near the smooth muscle cells of the vasculature (i.e., renal artery, AA, EA) & the proximal tubules.
- 3 major effects:
    - Vasoconstriction.
    - Enhance Na^+ reabsorption by proximal tubule cells.
    - Stimulate renin secretion (GC of the AA).

Afferent Innervation

- Renal nerves also include afferent (i.e., sensory) fibers
- Detects baroreceptor and chemoreceptor impulses.
- Baroreceptors: Detect changes in perfusion pressure in the interlobular arteries & AA.
- Chemoreceptors: Detect elevated E.C. $[K^+]$ & $[H^+]$ in the renal pelvis



Renal Efferents



Sympathetic Nervous system

NE	NE	NE
↓ Arteries Arterioles	↓ Proximal Tubules	↓ JGA
↓ SMCs	↓ Proximal Tubular cells	↓ Granular cells
↓ α_1 rec	↓ α_1/α_2 rec	↓ β_1 rec
↓ Vasoconstriction	↓ Na^+ reabsorption	↓ Renin release

Note: Intrarenal production of dopamine is not regulated by renal sympathetic nerve activity,

Control of Bladder Tone

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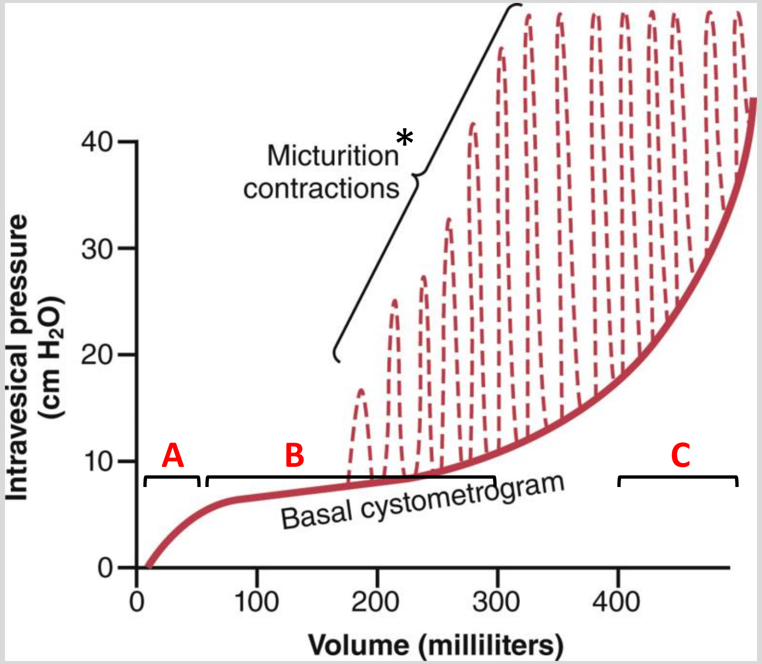
- Bladder tone is defined by the relationship between bladder volume & internal (intravesical) pressure & can be evaluated by a cystometrogram.

A: Increase in bladder volume from 0-~50 mL produces a moderately steep increase in pressure.

B: Volume increases (to ~300 mL) produce almost no pressure increase, reflecting relaxation of bladder smooth muscle.

C: At volumes >400 mL, additional increases in volume produce steep increments in pressure. Bladder tone, up to the point of triggering the micturition reflex, is independent of extrinsic bladder innervation.

- The micturition reflex is normally inhibited by the pontine micturition center (PMC).
- The urge for voluntary bladder emptying starts ~150 mL
- Senses fullness at 400-500 mL
- Voiding phase:
  - Begins with a voluntary relaxation of the ES followed by relaxation of the IS
  - Afferents signal are sent when a small amount of urine reaches the proximal (posterior) urethra.
  - The micturition reflex: the PMC no longer inhibit the parasympathetic preganglionic neurons that innervate the detrusor muscle.
  - Result: The bladder contracts, expelling urine.
  - The initial bladder contractions → Sensory impulses from stretch receptors & establishing a self-regenerating process



Cystometrogram (CMG):

- Diagnostic test that measures bladder pressure, capacity & muscle function
- Via insertion of a catheter through the urethra & emptying the bladder, and then recording the pressure while filling the bladder with 50-mL increments of water

*Micturition = urination

PMC: Cortical and suprapontine centers in the brain